

Problem 2

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Consider the following signalling game. The current owners of the Firm ABC inc. and an investor are in negotiations. The investor would like to enter as a shareholder of the firm but he is uncertain of whether the firm is in financial distress or not. On the other hand, the owners of the firm know exactly its financial condition. The investor has a prior belief about whether the Firm is in good standing: there is probability $\frac{2}{3}$ that the Firm is in good condition. All of this is common knowledge.

If the firm is in good condition, its profits will be equal to 8 Million, whereas if the firm is in bad condition, its profit will be equal to 4 Million. The investor thinks of entering as a 25% shareholder, and thus having the respective share in the profits. The cost of buying this share is 1 million.

The current owners of the firm are considering ways of signaling the firm's financial situation. They can pay a high or a low dividend. A high dividend will involve returning 75% of the profits to the shareholders, and a low dividend 40% of the profits.

1. Draw the game tree that corresponds to this problem.
Hint: when a firm is in good standing and issues dividends, the current owners receive either $0.75 \cdot 0.75 \cdot 8 + 1 = 5.5$ Million or $0.75 \cdot 8 = 6$ Million in payoffs depending on whether the investor buys equity or not.
2. Describe all the pure-strategy pooling and separating Perfect Bayesian Equilibria.

1.
 $share \cdot pct\ of\ profit \cdot profit$

And then ± 1 since the cost of buying the share is 1 million

π_H, H, A

$$u_1 = .75 \cdot .75 \cdot 8 + 1$$
$$u_I = 5.5000 \tag{1.1.1}$$

$$u_2 = .25 \cdot .75 \cdot 8 - 1$$
$$u_2 = 0.5000 \tag{1.1.2}$$

π_H, H, R

$$u_1 = 1 \cdot .75 \cdot 8$$
$$u_I = 6.00 \tag{1.1.3}$$
$$u_2 = 0$$

$$\boldsymbol{\pi}_H, L, A$$

$$u_1 = .4 \cdot .75 \cdot 8 + 1$$

$$u_I = 3.400 \tag{1.1.4}$$

$$u_2 = .25 \cdot .4 \cdot 8 - 1$$

$$u_2 = -0.200 \tag{1.1.5}$$

$$\boldsymbol{\pi}_H, L, R$$

$$u_1 = .4 \cdot 8$$

$$u_I = 3.2 \tag{1.1.6}$$

$$u_2 = 0$$

$$\pi_L, H, A$$

$$u_1 = .75 \cdot .75 \cdot 4 + 1$$

$$u_I = 3.2500 \tag{1.1.7}$$

$$u_2 = .25 \cdot .75 \cdot 4 - 1$$

$$u_2 = -0.2500 \tag{1.1.8}$$

$$\pi_L, H, R$$

$$u_1 = 1 \cdot .75 \cdot 4$$

$$u_I = 3.00 \tag{1.1.9}$$

$$u_2 = 0$$

$$u_2 = 0 \tag{1.1.10}$$

$$\boldsymbol{\pi}_L, L, A$$

$$u_1 = .75 \cdot .4 \cdot 4 + 1$$

$$u_I = 2.200 \tag{1.1.11}$$

$$u_2 = .25 \cdot .4 \cdot 4 - 1$$

$$u_2 = -0.600 \tag{1.1.12}$$

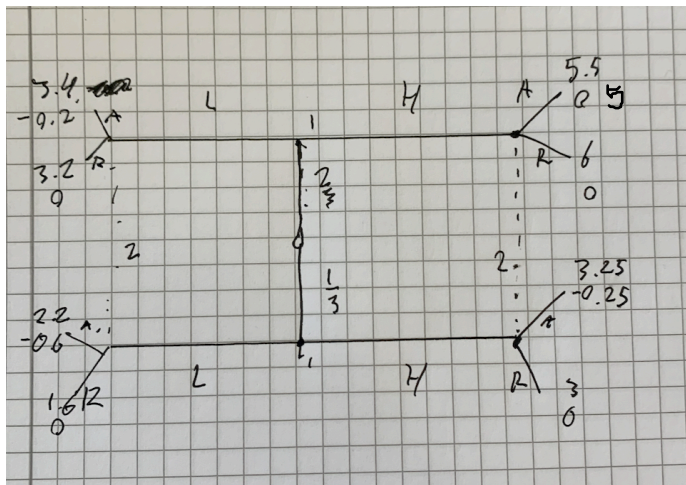
$$\boldsymbol{\pi}_L, L, R$$

$$u_1 = 1 \cdot .4 \cdot 4$$

$$u_I = 1.6 \tag{1.1.13}$$

$$u_2 = 0$$

$$u_2 = 0 \tag{1.1.14}$$



2.

α states the beliefs in low dividend scenario, left part

β the beliefs in high dividend, right part

I start with the possible pooling equilibria:

LL

I calculate expected utility:

$$EU_2[A|\alpha] = -0.2\left(\frac{2}{3}\right) - 0.6\left(\frac{1}{3}\right) = -\frac{1}{3}$$

$$EU_2[R|\alpha] = 0\left(\frac{2}{3}\right) + 0\left(\frac{1}{3}\right) = 0$$

Implying that $EU_2[R|\alpha] > EU_2[A|\alpha]$

But both payoffs in H in the good state — upper half — is better than the outcome for player 1 in L, i.e., $5.5 > 3.2$ and $6 > 3.2$

HH

I calculate expected utility:

$$EU_2 \left[A \mid \beta \right] = 0.5 \left(\frac{2}{3} \right) - 0.25 \cdot \left(\frac{1}{3} \right) = 0.25$$

$$EU_2 \left[R \mid \beta \right] = 0 \cdot \left(\frac{2}{3} \right) + 0 \cdot \left(\frac{1}{3} \right) = 0$$

Implying that

$$EU_2 \left[R \mid \beta \right] < EU_2 \left[A \mid \beta \right]$$

It is clear to see that player 1 will never deviate because in the upper half: $6 > 3.4$ and in the lower half: $3 > 1.6$

Therefore, this is a SPNE:

$$(HH, RA \text{ for } \alpha \in [0, 1], \beta = \frac{2}{3})$$

HL

It is clear to see that HL is not a SPNE: Player 1 will deviate in the lower and upper half because this will give him a payoff of $3.25 > 1.6$ and $5.5 > 3.2$

LH

This is not an equilibrium either. In the upper half, player 1 will deviate because $6 > 3.2$ and $3 > 1.6$

